

Example 1: Sweet Apple Acres

Applejack has come to us: “Between the Red Delicious row, the Honeycrisp row, and that new Zap Apple row Granny Smith insisted on plantin’, I reckon one of ‘em oughta be doin’ better than the others. Can you tell me if they’re really any different, or am I just imaginin’ things?”

- Glimpse of data:

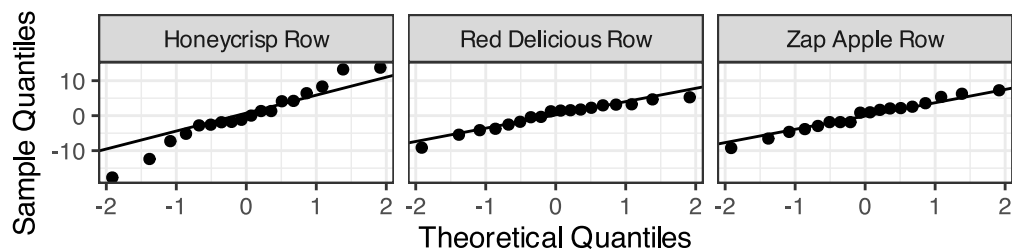
```
# A tibble: 3 × 2
  row      bushels
<chr>    <dbl>
1 Red Delicious Row  42.2
2 Red Delicious Row  45.3
3 Red Delicious Row  42.1
```

- Checking the normality of our Sweet Apple Acres dataset,

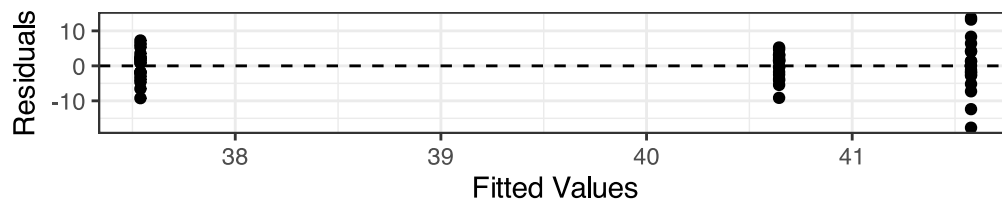
```
apple_harvest |> anova_check(outcome = bushels,
                             grouping = row)
```

One-Way ANOVA Assumptions

QQ Plots of Residuals by Group



Residuals vs Fitted Values



- Constructing the ANOVA table,

```
apple_harvest |> anovas(outcome = bushels,
                        grouping = row)
```

One-way ANOVA

Source	df	SS	MS	F	p-value
row	2	160.9693	80.4846	2.468	0.095
Residuals	51	1663.4383	32.6164		

Hypotheses (row):

H_0 : All group means are equal (no effect of row)

H_1 : At least one group mean differs (effect of row exists)

Test statistic: $F(2, 51) = 2.468$

p-value: 0.095

Conclusion: Fail to reject the null hypothesis ($p = 0.095 \geq \alpha = 0.050$)

- Applying the Kruskal-Wallis,

```
apple_harvest |> kruskal_wallis(outcome = bushels,
                              grouping = row)
```

Kruskal-Wallis test

$M[\text{Honeycrisp Row}] = 41$ ($n = 18$)

$M[\text{Red Delicious Row}] = 42$ ($n = 18$)

$M[\text{Zap Apple Row}] = 38.45$ ($n = 18$)

Hypotheses:

H_0 : All group distributions are equal (no effect of row)

H_1 : At least one group distribution differs in location (effect of row exists)

Test statistic: $H(2) = 5.953$

p-value: 0.051

Conclusion: Fail to reject the null hypothesis ($p = 0.051 \geq \alpha = 0.050$)

Q: Should we perform posthoc testing?

Example 2: Rarity's Boutique

- Rarity flits between racks of fabric: “Darling, I need your professional opinion. Between my Canterlot Couture line, my everyday Ponyville Casual line, and my absolutely *dazzling* Gala Elegance line, I use wildly different amounts of gemstones. Is the difference really as dramatic as I think?”

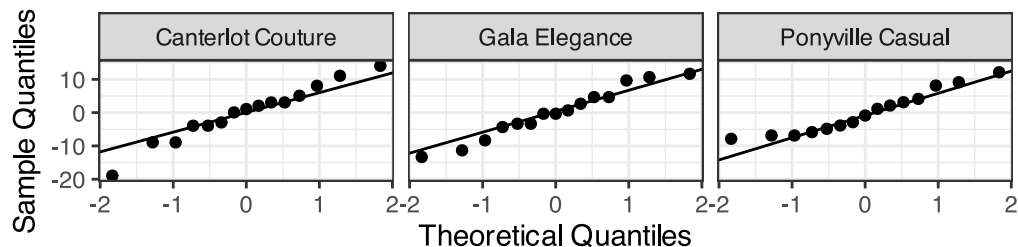
```
# A tibble: 3 × 2
  collection      gem_count
  <chr>          <dbl>
1 Canterlot Couture    59
2 Canterlot Couture    57
3 Canterlot Couture    64
```

- Checking the normality of Rarity's dataset,

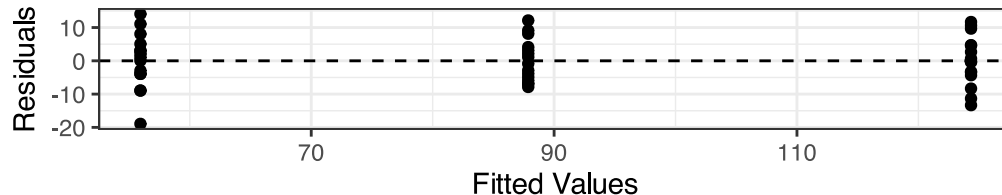
```
boutique_gems |> anova_check(outcome = gem_count,
                             grouping = collection)
```

One-Way ANOVA Assumptions

QQ Plots of Residuals by Group



Residuals vs Fitted Values



- Constructing the ANOVA table,

```
boutique_gems |> anovas(outcome = gem_count,
                        grouping = collection)
```

One-way ANOVA

Source	df	SS	MS	F	p-value
collection	2	35140.5778	17570.2889	311.110	< 0.001
Residuals	42	2372.0000	56.4762		

Hypotheses (collection):

H_0 : All group means are equal (no effect of collection)

H_1 : At least one group mean differs (effect of collection exists)

Test statistic: $F(2, 42) = 311.11$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

- Applying the Kruskal-Wallis,

```
boutique_gems |> kruskal_wallis(outcome = gem_count,
                                grouping = collection)
```

Kruskal-Wallis test

M[Canterlot Couture] = 57 (n = 15)

M[Gala Elegance] = 124 (n = 15)

M[Ponyville Casual] = 87 (n = 15)

Hypotheses:

H_0 : All group distributions are equal (no effect of collection)

H_1 : At least one group distribution differs in location (effect of collection exists)

Test statistic: $H(2) = 39.148$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

Q: Should we perform posthoc testing?

Q: Treat this as an exploratory study and use the following output to determine which pairwise differences exist. Circle the output you are interpreting.

- Posthoc tests:

```
boutique_gems |> posthoc(outcome = gem_count,
                        grouping = collection,
                        adjust = TRUE,
                        nonparametric = FALSE)
```

Post-hoc analysis: Tukey's HSD

Following: one-way ANOVA (gem_count ~ collection)

Comparison	Difference	p-value	
Gala Elegance vs Canterlot Couture	68.4000	< 0.001	*
Ponyville Casual vs Canterlot Couture	31.9333	< 0.001	*
Ponyville Casual vs Gala Elegance	-36.4667	< 0.001	*

* $p < \alpha = 0.050$

```
boutique_gems |> posthoc(outcome = gem_count,
                        grouping = collection,
                        adjust = FALSE,
                        nonparametric = FALSE)
```

Post-hoc analysis: Fisher's LSD

Following: one-way ANOVA (gem_count ~ collection)

Comparison	Difference	p-value	
Gala Elegance vs Canterlot Couture	68.4000	< 0.001	*
Ponyville Casual vs Canterlot Couture	31.9333	< 0.001	*
Ponyville Casual vs Gala Elegance	-36.4667	< 0.001	*

* $p < \alpha = 0.050$

```
boutique_gems |> posthoc(outcome = gem_count,
                        grouping = collection,
                        adjust = TRUE,
                        nonparametric = TRUE)
```

Post-hoc analysis: Dunn's test (Bonferroni adjustment)

Following: Kruskal-Wallis (gem_count ~ collection)

Comparison	z	p-value	
Canterlot Couture vs Gala Elegance	-6.257	< 0.001	*
Canterlot Couture vs Ponyville Casual	-3.128	0.003	*
Gala Elegance vs Ponyville Casual	3.128	0.003	*

* $p < \alpha = 0.050$

```
boutique_gems |> posthoc(outcome = gem_count,  
                        grouping = collection,  
                        adjust = FALSE,  
                        nonparametric = TRUE)
```

Post-hoc analysis: Dunn's test (no adjustment)

Following: Kruskal-Wallis (gem_count ~ collection)

Comparison	z	p-value	
Canterlot Couture vs Gala Elegance	-6.257	< 0.001	*
Canterlot Couture vs Ponyville Casual	-3.128	< 0.001	*
Gala Elegance vs Ponyville Casual	3.128	< 0.001	*

* $p < \alpha = 0.050$

Example 3: Ponyville Diner

- Sunny Skies wipes down the counter after the lunch rush: “Feels like Caramel always walks away with better tips than me and Lyra put together. Before I start complainin’ to the manager, can you check if there’s actually a difference in how much we’re bringing in?”

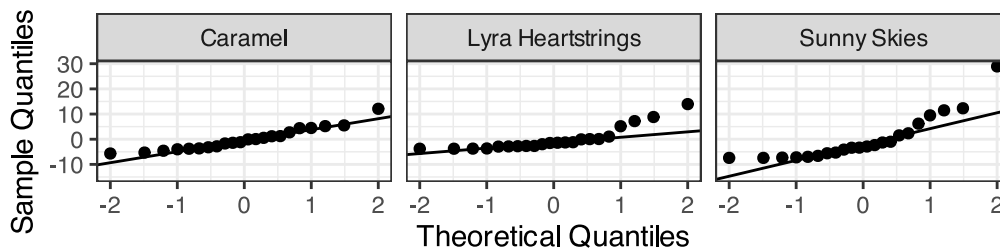
```
# A tibble: 3 × 2
  server    tip_bits
  <chr>    <dbl>
1 Sunny Skies    4.08
2 Sunny Skies    6.12
3 Sunny Skies    0.12
```

- Checking the normality of the Ponyville Diner dataset,

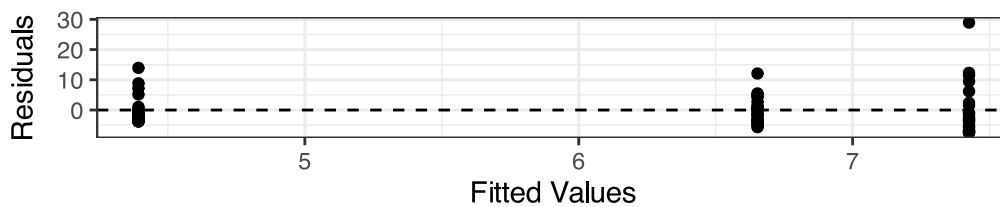
```
ponyville_diner_tips |> anova_check(outcome = tip_bits,
                                   grouping = server)
```

One-Way ANOVA Assumptions

QQ Plots of Residuals by Group



Residuals vs Fitted Values



- Constructing the ANOVA table,

```
ponyville_diner_tips |> anovas(outcome = tip_bits,
                               grouping = server)
```

One-way ANOVA

Source	df	SS	MS	F	p-value
--------	----	----	----	---	---------

server	2	109.2923	54.6461	1.356	0.265
Residuals	63	2539.3768	40.3076		

Hypotheses (server):

H_0 : All group means are equal (no effect of server)

H_1 : At least one group mean differs (effect of server exists)

Test statistic: $F(2, 63) = 1.356$

p-value: 0.265

Conclusion: Fail to reject the null hypothesis ($p = 0.265 \geq \alpha = 0.050$)

- Applying the Kruskal-Wallis,

```
ponyville_diner_tips |> kruskal_wallis(outcome = tip_bits,
                                     grouping = server)
```

Kruskal-Wallis test

$M[\text{Caramel}] = 6.03$ ($n = 22$)

$M[\text{Lyra Heartstrings}] = 2.94$ ($n = 22$)

$M[\text{Sunny Skies}] = 4.305$ ($n = 22$)

Hypotheses:

H_0 : All group distributions are equal (no effect of server)

H_1 : At least one group distribution differs in location (effect of server exists)

Test statistic: $H(2) = 4.235$

p-value: 0.120

Conclusion: Fail to reject the null hypothesis ($p = 0.120 \geq \alpha = 0.050$)

Q: Should we perform posthoc testing?

Let's Practice! 1: Wonderbolts Academy

- Fleetfoot drops a stack of stopwatch printouts on the table: "Command wants to know if lap times actually differ across our four training squads: Thunderhead, Stormwing, Skybolt, and Downburst."

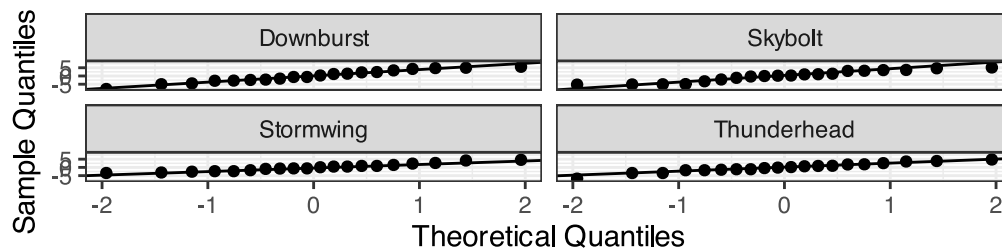
```
# A tibble: 3 × 2
  squad    lap_seconds
<chr>    <dbl>
1 Thunderhead    60.4
2 Thunderhead    59.6
3 Thunderhead    62
```

- Checking the normality,

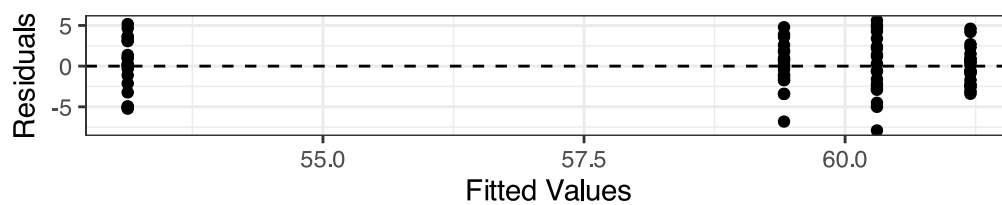
```
wonderbolts_laptimes |> anova_check(outcome = lap_seconds,
                                     grouping = squad)
```

One-Way ANOVA Assumptions

QQ Plots of Residuals by Group



Residuals vs Fitted Values



- Constructing the ANOVA table,

```
wonderbolts_laptimes |> anovas(outcome = lap_seconds,
                                grouping = squad)
```

One-way ANOVA

Source	df	SS	MS	F	p-value
--------	----	----	----	---	---------

squad	3	804.4305	268.1435	28.098	< 0.001
Residuals	76	725.2770	9.5431		

Hypotheses (squad):

H_0 : All group means are equal (no effect of squad)

H_1 : At least one group mean differs (effect of squad exists)

Test statistic: $F(3, 76) = 28.098$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

- Applying the Kruskal-Wallis,

```
wonderbolts_laptimes |> kruskal_wallis(outcome = lap_seconds,
                                         grouping = squad)
```

Kruskal-Wallis test

$M[\text{Downburst}] = 60.15$ ($n = 20$)

$M[\text{Skybolt}] = 53.35$ ($n = 20$)

$M[\text{Stormwing}] = 60.95$ ($n = 20$)

$M[\text{Thunderhead}] = 59.45$ ($n = 20$)

Hypotheses:

H_0 : All group distributions are equal (no effect of squad)

H_1 : At least one group distribution differs in location (effect of squad exists)

Test statistic: $H(3) = 38.039$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

Q: Should we perform posthoc testing?

Q: Treat this as a confirmatory study and use the following output to determine which pairwise differences exist. Circle the output you are interpreting.

- Posthoc tests:

```
wonderbolts_laptimes |> posthoc(outcome = lap_seconds,
                                grouping = squad,
                                adjust = TRUE,
                                nonparametric = FALSE)
```

Post-hoc analysis: Tukey's HSD

Following: one-way ANOVA (lap_seconds ~ squad)

Comparison	Difference	p-value	
Skybolt vs Downburst	-7.1750	< 0.001	*
Stormwing vs Downburst	0.8950	0.796	
Thunderhead vs Downburst	-0.8900	0.799	
Stormwing vs Skybolt	8.0700	< 0.001	*
Thunderhead vs Skybolt	6.2850	< 0.001	*
Thunderhead vs Stormwing	-1.7850	0.269	

* $p < \alpha = 0.050$

```
wonderbolts_laptimes |> posthoc(outcome = lap_seconds,
                                grouping = squad,
                                adjust = FALSE,
                                nonparametric = FALSE)
```

Post-hoc analysis: Fisher's LSD

Following: one-way ANOVA (lap_seconds ~ squad)

Comparison	Difference	p-value	
Skybolt vs Downburst	-7.1750	< 0.001	*
Stormwing vs Downburst	0.8950	0.362	
Stormwing vs Skybolt	8.0700	< 0.001	*
Thunderhead vs Downburst	-0.8900	0.365	
Thunderhead vs Skybolt	6.2850	< 0.001	*
Thunderhead vs Stormwing	-1.7850	0.072	

* $p < \alpha = 0.050$

```
wonderbolts_laptimes |> posthoc(outcome = lap_seconds,
                                grouping = squad,
                                adjust = TRUE,
                                nonparametric = TRUE)
```

Post-hoc analysis: Dunn's test (Bonferroni adjustment)

Following: Kruskal-Wallis (lap_seconds ~ squad)

Comparison	z	p-value	
Downburst vs Skybolt	4.787	< 0.001	*
Downburst vs Stormwing	-0.915	1.000	
Skybolt vs Stormwing	-5.703	< 0.001	*
Downburst vs Thunderhead	0.701	1.000	
Skybolt vs Thunderhead	-4.086	< 0.001	*
Stormwing vs Thunderhead	1.616	0.318	

* $p < \alpha = 0.050$

```
wonderbolts_laptimes |> posthoc(outcome = lap_seconds,
                                grouping = squad,
                                adjust = FALSE,
                                nonparametric = TRUE)
```

Post-hoc analysis: Dunn's test (no adjustment)

Following: Kruskal-Wallis (lap_seconds ~ squad)

Comparison	z	p-value	
Downburst vs Skybolt	4.787	< 0.001	*
Downburst vs Stormwing	-0.915	0.180	
Skybolt vs Stormwing	-5.703	< 0.001	*
Downburst vs Thunderhead	0.701	0.242	
Skybolt vs Thunderhead	-4.086	< 0.001	*
Stormwing vs Thunderhead	1.616	0.053	

* $p < \alpha = 0.050$

Let's Practice! 2: Klugetown Bazaar

- A weary trader waves you over between haggling matches: “Rustbolt, Glimmering, and Junker each swear their stall pulls in the best prices. I’ve got a season’s worth of sale records — do the three stalls actually sell for different amounts, or is everyone just talkin’ up their own wares?”

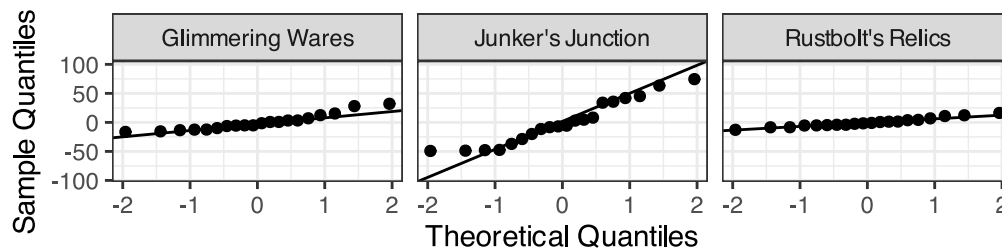
```
# A tibble: 3 × 2
  stall      sale_price_bits
  <chr>      <dbl>
1 Rustbolt's Relics      21.2
2 Rustbolt's Relics      18.9
3 Rustbolt's Relics      26.8
```

- Checking the normality,

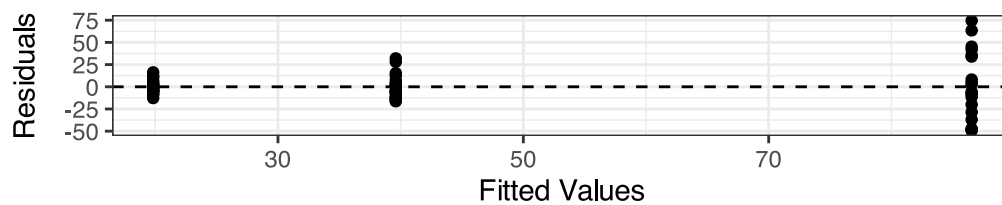
```
klugetown_bazaar |> anova_check(outcome = sale_price_bits,
                                grouping = stall)
```

One-Way ANOVA Assumptions

QQ Plots of Residuals by Group



Residuals vs Fitted Values



- Constructing the ANOVA table,

```
klugetown_bazaar |> anovas(outcome = sale_price_bits,
                             grouping = stall)
```

One-way ANOVA

Source	df	SS	MS	F	p-value
--------	----	----	----	---	---------

stall	2	46965.2768	23482.6384	40.982	< 0.001
Residuals	57	32661.2297	573.0040		

Hypotheses (stall):

H_0 : All group means are equal (no effect of stall)

H_1 : At least one group mean differs (effect of stall exists)

Test statistic: $F(2, 57) = 40.982$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

- Applying the Kruskal-Wallis,

```
klugetown_bazaar |> kruskal_wallis(outcome = sale_price_bits,
                                   grouping = stall)
```

Kruskal-Wallis test

$M[\text{Glimmering Wares}] = 36.36$ ($n = 20$)

$M[\text{Junker's Junction}] = 80.29$ ($n = 20$)

$M[\text{Rustbolt's Relics}] = 18.565$ ($n = 20$)

Hypotheses:

H_0 : All group distributions are equal (no effect of stall)

H_1 : At least one group distribution differs in location (effect of stall exists)

Test statistic: $H(2) = 43.348$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

Q: Should we perform posthoc testing?

Q: Treat this as a confirmatory study and use the following output to determine which pairwise differences exist. Circle the output you are interpreting.

- Posthoc tests:

```
klugetown_bazaar |> posthoc(outcome = sale_price_bits,
                             grouping = stall,
                             adjust = TRUE,
                             nonparametric = FALSE)
```

Post-hoc analysis: Tukey's HSD

Following: one-way ANOVA (sale_price_bits ~ stall)

Comparison	Difference	p-value	
Junker's Junction vs Glimmering Wares	46.9385	< 0.001	*
Rustbolt's Relics vs Glimmering Wares	-19.7740	0.030	*
Rustbolt's Relics vs Junker's Junction	-66.7125	< 0.001	*

* $p < \alpha = 0.050$

```
klugetown_bazaar |> posthoc(outcome = sale_price_bits,
                             grouping = stall,
                             adjust = FALSE,
                             nonparametric = FALSE)
```

Post-hoc analysis: Fisher's LSD

Following: one-way ANOVA (sale_price_bits ~ stall)

Comparison	Difference	p-value	
Junker's Junction vs Glimmering Wares	46.9385	< 0.001	*
Rustbolt's Relics vs Glimmering Wares	-19.7740	0.011	*
Rustbolt's Relics vs Junker's Junction	-66.7125	< 0.001	*

* $p < \alpha = 0.050$

```
klugetown_bazaar |> posthoc(outcome = sale_price_bits,
                             grouping = stall,
                             adjust = TRUE,
                             nonparametric = TRUE)
```

Post-hoc analysis: Dunn's test (Bonferroni adjustment)

Following: Kruskal-Wallis (sale_price_bits ~ stall)

Comparison	z	p-value	
Glimmering Wares vs Junker's Junction	-3.038	0.004	*
Glimmering Wares vs Rustbolt's Relics	3.540	< 0.001	*
Junker's Junction vs Rustbolt's Relics	6.578	< 0.001	*

* $p < \alpha = 0.050$

```
klugetown_bazaar |> posthoc(outcome = sale_price_bits,  
                             grouping = stall,  
                             adjust = FALSE,  
                             nonparametric = TRUE)
```

Post-hoc analysis: Dunn's test (no adjustment)

Following: Kruskal-Wallis (sale_price_bits ~ stall)

Comparison	z	p-value	
Glimmering Wares vs Junker's Junction	-3.038	0.001	*
Glimmering Wares vs Rustbolt's Relics	3.540	< 0.001	*
Junker's Junction vs Rustbolt's Relics	6.578	< 0.001	*

* $p < \alpha = 0.050$

Let's Practice! 3: Baltimore Harbor

- The harbormaster spreads a stack of logbooks across the table: “Four dock crews unload every ship that comes through: Anchor, Tidewater, Saltmarsh, and Driftwood. Some of the sailors have been grumbling that one crew is dragging everyone’s schedule behind. Before I say anything to the crews, I want to know if unloading time actually differs across them.”

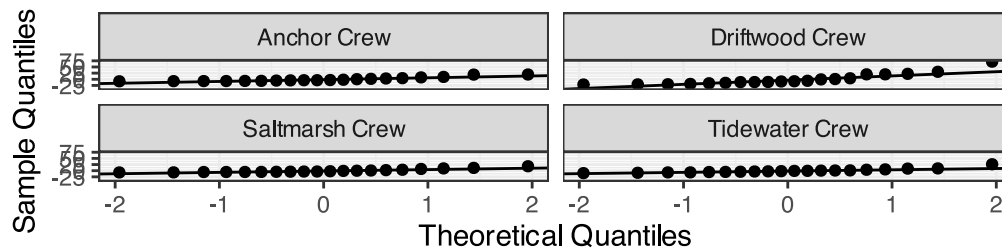
```
# A tibble: 3 × 2
  crew      unload_minutes
<chr>      <dbl>
1 Anchor Crew      1.3
2 Anchor Crew      2.19
3 Anchor Crew      5.13
```

- Checking the normality,

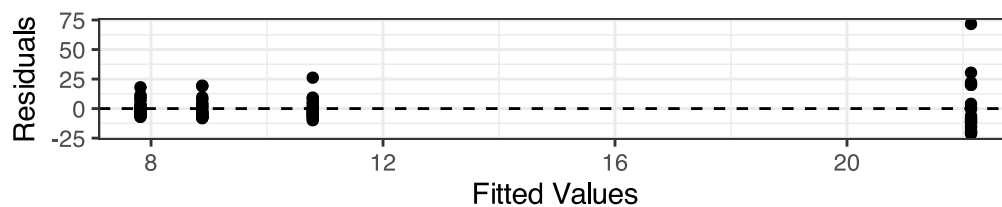
```
baltimore_harbor |> anova_check(outcome = unload_minutes,
                                grouping = crew)
```

One-Way ANOVA Assumptions

QQ Plots of Residuals by Group



Residuals vs Fitted Values



- Constructing the ANOVA table,

```
baltimore_harbor |> anovas(outcome = unload_minutes,
                           grouping = crew)
```

One-way ANOVA

Source	df	SS	MS	F	p-value
crew	3	2616.1986	872.0662	4.906	0.004
Residuals	76	13509.4136	177.7554		

Hypotheses (crew):

H_0 : All group means are equal (no effect of crew)

H_1 : At least one group mean differs (effect of crew exists)

Test statistic: $F(3, 76) = 4.906$

p-value: 0.004

Conclusion: Reject the null hypothesis ($p = 0.004 < \alpha = 0.050$)

- Applying the Kruskal-Wallis,

```
baltimore_harbor |> kruskal_wallis(outcome = unload_minutes,
                                   grouping = crew)
```

Kruskal-Wallis test

$M[\text{Anchor Crew}] = 5.44$ ($n = 20$)

$M[\text{Driftwood Crew}] = 13.67$ ($n = 20$)

$M[\text{Saltmarsh Crew}] = 5.09$ ($n = 20$)

$M[\text{Tidewater Crew}] = 9.475$ ($n = 20$)

Hypotheses:

H_0 : All group distributions are equal (no effect of crew)

H_1 : At least one group distribution differs in location (effect of crew exists)

Test statistic: $H(3) = 7.917$

p-value: 0.048

Conclusion: Reject the null hypothesis ($p = 0.048 < \alpha = 0.050$)

Q: Should we perform posthoc testing?

Q: Treat this as a confirmatory study and use the following output to determine which pairwise differences exist. Circle the output you are interpreting.

- Posthoc tests:

```
baltimore_harbor |> posthoc(outcome = unload_minutes,
                             grouping = crew,
                             adjust = TRUE,
                             nonparametric = FALSE)
```

Post-hoc analysis: Tukey's HSD

Following: one-way ANOVA (unload_minutes ~ crew)

Comparison	Difference	p-value
Driftwood Crew vs Anchor Crew	13.2555	0.012 *
Saltmarsh Crew vs Anchor Crew	-1.0660	0.994
Tidewater Crew vs Anchor Crew	1.9050	0.969
Saltmarsh Crew vs Driftwood Crew	-14.3215	0.006 *
Tidewater Crew vs Driftwood Crew	-11.3505	0.042 *
Tidewater Crew vs Saltmarsh Crew	2.9710	0.895

* $p < \alpha = 0.050$

```
baltimore_harbor |> posthoc(outcome = unload_minutes,
                             grouping = crew,
                             adjust = FALSE,
                             nonparametric = FALSE)
```

Post-hoc analysis: Fisher's LSD

Following: one-way ANOVA (unload_minutes ~ crew)

Comparison	Difference	p-value
Driftwood Crew vs Anchor Crew	13.2555	0.002 *
Saltmarsh Crew vs Anchor Crew	-1.0660	0.801
Saltmarsh Crew vs Driftwood Crew	-14.3215	0.001 *
Tidewater Crew vs Anchor Crew	1.9050	0.653
Tidewater Crew vs Driftwood Crew	-11.3505	0.009 *
Tidewater Crew vs Saltmarsh Crew	2.9710	0.483

* $p < \alpha = 0.050$

```
baltimore_harbor |> posthoc(outcome = unload_minutes,
                           grouping = crew,
                           adjust = TRUE,
                           nonparametric = TRUE)
```

Post-hoc analysis: Dunn's test (Bonferroni adjustment)

Following: Kruskal-Wallis (unload_minutes ~ crew)

Comparison	z	p-value
Anchor Crew vs Driftwood Crew	-2.286	0.067
Anchor Crew vs Saltmarsh Crew	0.214	1.000
Driftwood Crew vs Saltmarsh Crew	2.501	0.037 *
Anchor Crew vs Tidewater Crew	-1.085	0.833
Driftwood Crew vs Tidewater Crew	1.201	0.689
Saltmarsh Crew vs Tidewater Crew	-1.300	0.581

* $p < \alpha = 0.050$

```
baltimore_harbor |> posthoc(outcome = unload_minutes,
                           grouping = crew,
                           adjust = FALSE,
                           nonparametric = TRUE)
```

Post-hoc analysis: Dunn's test (no adjustment)

Following: Kruskal-Wallis (unload_minutes ~ crew)

Comparison	z	p-value
Anchor Crew vs Driftwood Crew	-2.286	0.011 *
Anchor Crew vs Saltmarsh Crew	0.214	0.415
Driftwood Crew vs Saltmarsh Crew	2.501	0.006 *
Anchor Crew vs Tidewater Crew	-1.085	0.139
Driftwood Crew vs Tidewater Crew	1.201	0.115
Saltmarsh Crew vs Tidewater Crew	-1.300	0.097

* $p < \alpha = 0.050$